HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Mapped Continued Process Verification (CPV) data requirements to source systems including **MES/eBR**, **process historian**, and **ERP** for a global pharmaceutical manufacturer.
- Assessed manufacturing data completeness and accuracy, logging and prioritising gaps for remediation activities such as data cleansing and backfilling from source records.
- Prepared and cleansed batch, phase, and equipment hierarchy data for loading into a site's **digital trending platform**, ensuring adherence to GMP data integrity standards.
- Developed Python scripts to automate data extraction and transformation processes, reducing manual effort by approximately **40%** for batch record analysis.
- Implemented data validation checks using **SQL** across **PostgreSQL** and **Oracle** databases to ensure data integrity and accuracy for regulatory reporting.
- Coordinated daily data remediation tasks for two junior CPV Production Support consultants, providing guidance on data quality standards and remediation approaches.
- Containerised data processing applications using **Docker** and deployed them to **AWS EC2** instances, improving deployment consistency and reducing setup time by **30%**.

SKILLS

Manufacturing & Data Integrity – Continued Process Verification (CPV), MES/eBR, Process Historian, Data Lakes, Data Integrity (ALCOA+), EU GMP Annex 11, 21 CFR Part 11, Computerised System Validation, Batch Data, ISA-88, ISA-95

Data Analysis & BI – SQL, Python, Power BI, Data Cleansing, Data Backfill, Data Modelling, Data Warehousing, ETL, Data Visualisation

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Programming Languages – Python, SQL, Java, Spring Boot

Databases – PostgreSQL, Oracle, MySQL

Tools & Platforms – AVEVA, OSIsoft PI, Jira, Confluence, Splunk, Dynatrace

PROJECTS

Manufacturing Data Remediation Platform (HCL Technologies)

- Developed a Python-based application to ingest and validate manufacturing data from **MES** and **ERP** systems, flagging data integrity issues for review.
- Designed and implemented **SQL** scripts for data cleansing and backfilling operations, processing over **50,000** batch records to meet CPV requirements.
- Created interactive **Power BI** dashboards to visualise data quality metrics and track remediation progress for key manufacturing processes.

Automated Data Pipeline for Process Historian (HCL Technologies)

- Built automated data extraction pipelines using **Python** and **SQL** to pull time-series data from a **process historian** database.
- Transformed and standardised raw process data, applying unit conversions and aligning timestamps to support downstream analysis and trending.
- Deployed the pipeline on **AWS Lambda**, processing approximately **2,000** data points per minute for a financial-services operations dashboard.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)